SIMATS ENGINEERING

ASSESSMENT TOOL I – RELATIONSHIP VISUALIZATION LAB

Course: Data Handling and Visualization	**Code: DSA0603**	**CO:** CO4
Duration: 60 min	**Total Marks:** 50	Reg No : 192524108 Name: S. LAKSHITA

COURSE OUTCOME COVERED

CO1 Understand the fundamental concepts, principles, and techniques of data visualization. (BL3)
Activity Tool : Relationship Visualization Lab
Weightage: 20%

Code of Conduct:

I S. LAKSHITA(192524108) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Selection of Relationship Visualization 25%	Relationship Analysis 25%	Application of Visualization Techniques 20%	Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Relationship Visualization Lab

Provided Datasets

The following files are supplied alongside this document so the lab can be completed without internet access to install extra data packages:

- mtcars.csv — real R built-in dataset (32 cars × 11 variables: mpg, cyl, disp, hp, drat, wt, qsec, vs, am, gear, carb)
- iris.csv — real R built-in dataset (150 flowers × 5 variables: Sepal.Length, Sepal.Width, Petal.Length, Petal.Width, Species)
- gapminder2007_practice.csv — illustrative practice dataset in the style of the Gapminder 2007 cross section (country, continent, year, gdpPercap, lifeExp, pop). Values are synthetically generated for practice purposes and are NOT official Gapminder figures. If internet access is available, students may instead run install.packages("gapminder") and use data(gapminder) for the authentic dataset.
- network_edges.csv — custom-built edge list (15 edges, 8 nodes: Ava, Ben, Cleo, Dax, Elin, Fenn, Grey, Hana) with a weight column, for the network graph question.

Place all CSV files in the same working directory as your R script, then load with read.csv("filename.csv").

Question 1: Scatterplot with Regression Diagnostics

Dataset provided: mtcars.csv

Using the mtcars dataset, create a scatterplot of mpg vs wt (weight) using ggplot2. Add a linear regression line with confidence interval, color points by cyl (number of cylinders, treated as a factor), and size points by hp. Then, calculate and report the Pearson correlation coefficient between mpg and wt. Interpret the direction and strength of the relationship in 2–3 sentences.

Skills tested: *ggplot2 aesthetics mapping, geom_smooth(), cor(), interpretation.*

Question 2: Correlogram Construction

Dataset provided: mtcars.csv (all numeric columns)

Using the mtcars dataset, compute the correlation matrix and visualize it as a correlogram using either the corrplot or GGally::ggcorr package. Configure the plot to:

- Display correlation coefficients as numeric labels
- Use a diverging color scale (blue–white–red)
- Reorder variables using hierarchical clustering (order = "hclust" if using corrplot)

Identify the two variable pairs with the strongest positive and strongest negative correlations. ***Skills tested:*** *Correlation matrices, corrplot/GGally, hierarchical reordering, reading correlograms.*

Question 3: Dimension Reduction Comparison (PCA vs. t-SNE)

Question 4: Bubble Chart for Multivariate Paired Data

Dataset provided: gapminder2007_practice.csv (or the gapminder package, year == 2007)

Filter for the year 2007 and create a bubble chart showing:

- x-axis: gdpPercap (log-transformed scale)
- y-axis: lifeExp
- bubble size: pop
- bubble color: continent

Add appropriate axis labels, a title, and use scale_size_continuous() to control bubble size range so points don't overplot excessively. Discuss one visible cluster or outlier in the data.

Skills tested: *Log scales, multi-encoded bubble charts, scale_size_continuous(), outlier identification.*

Question 5: Hexbin Plot vs. Network Graph

This question has two parts to compare visualization strategies for different relationship types.

Part a — Hexbin Plot

Dataset provided: Simulated in R (no file needed)

Simulate two correlated variables with 10,000 observations (set.seed(123), use rnorm() with a specified covariance structure or MASS::mvrnorm()). Create a hexbin plot using geom_hex() in ggplot2. Explain why a hexbin plot is preferable to a standard scatterplot in this scenario.

Part b — Network Graph

Dataset provided: network_edges.csv

Using the provided edge list (8 nodes, 15 weighted edges), build a network graph using igraph or ggraph. Size nodes by degree centrality and color edges by weight. Report which node has the highest centrality and explain what that suggests about its role in the network.

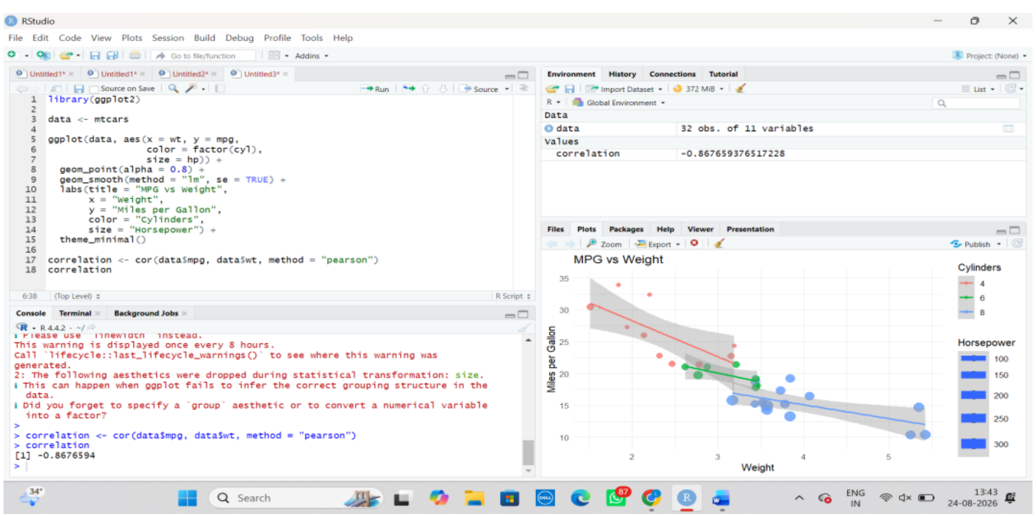
Skills tested: *geom_hex(), overplotting solutions, igraph/ggraph, centrality measures, network interpretation.*

ANSWERS

Q1. Scatterplot with Regression Diagnostics

R Code

```
library(ggplot2)

data <- read.csv("mtcars.csv")

ggplot(data, aes(x = wt, y = mpg,
                 color = factor(cyl), size = hp)) +
  geom_point(alpha = 0.8) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(title = "MPG vs Weight",
       x = "Weight",
       y = "Miles per Gallon",
       color = "Cylinders",
       size = "Horsepower") +
  theme_minimal()

correlation <- cor(data$mpg, data$wt, method = "pearson")
correlation
```



Result

Pearson correlation coefficient:

r ≈ -0.868

Interpretation

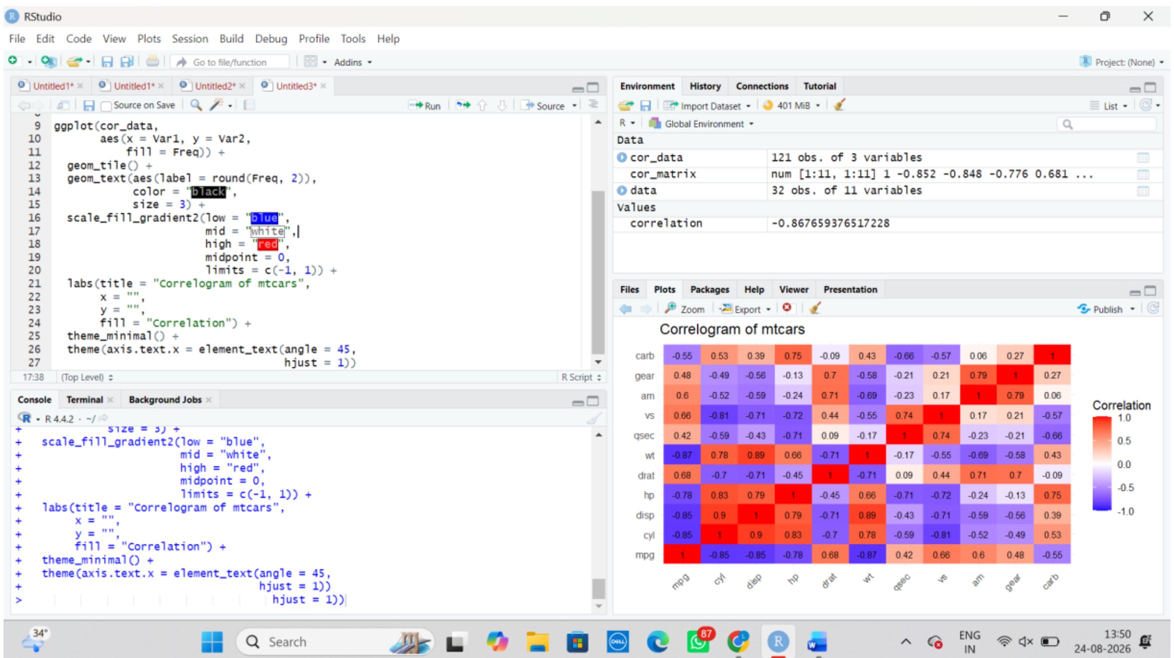
There is a **strong negative correlation** between mpg and wt. This means that as the weight of a car increases, its miles per gallon generally decreases. The relationship is strongly linear and negative.

Q2. Correlogram Construction

R Code

```
library(ggplot2)

data <- mtcars

cor_matrix <- cor(data)

cor_data <- as.data.frame(as.table(cor_matrix))

ggplot(cor_data,
       aes(x = Var1, y = Var2,
           fill = Freq)) +
  geom_tile() +
  geom_text(aes(label = round(Freq, 2)),
            color = "black",
            size = 3) +
  scale_fill_gradient2(low = "blue",
                       mid = "white",
                       high = "red",
                       midpoint = 0,
                       limits = c(-1, 1)) +
```

```
labs(title = "Correlogram of mtcars",
     x = "",
     y = "",
     fill = "Correlation") +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45,
                                 hjust = 1))
```



Strongest Positive Correlation

cyl and disp

Correlation:

r ≈ 0.902

This indicates a very strong positive relationship. Cars with more cylinders generally have larger engine displacement.

Strongest Negative Correlation

mpg and wt

Correlation:

r ≈ -0.868

This indicates a very strong negative relationship. Heavier cars generally have lower fuel efficiency.

Answer

- **Strongest positive:** cyl – disp → **0.902**
- **Strongest negative:** mpg – wt → **-0.868**

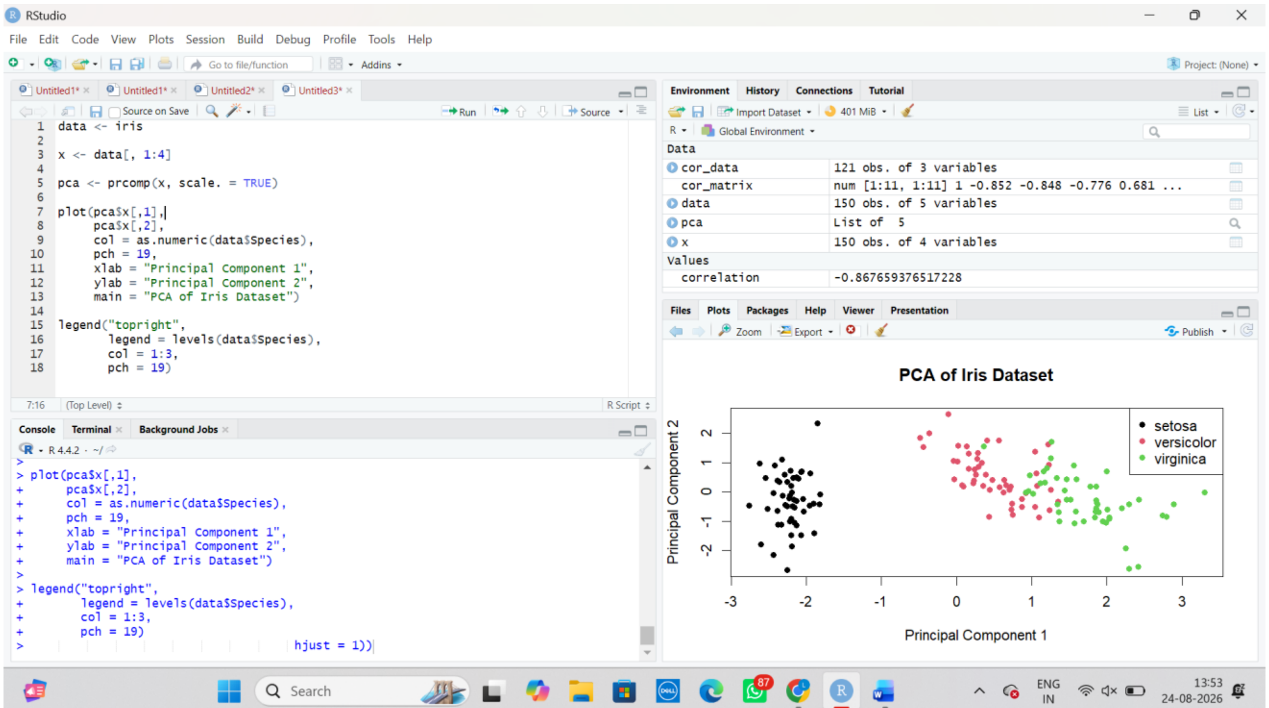
Q3. Dimension Reduction Comparison — PCA vs t-SNE

PCA	**t-SNE**
Principal Component Analysis	t-distributed Stochastic Neighbor Embedding
Linear dimensionality-reduction technique	Non-linear dimensionality-reduction technique
Preserves maximum variance	Focuses mainly on local neighbourhoods
Fast and relatively simple	Computationally more expensive
Components are interpretable	Axes usually have no direct meaning
Useful for overall structure	Useful for visualizing clusters

PCA

```
data <- iris

x <- data[, 1:4]

pca <- prcomp(x, scale. = TRUE)

plot(pca$x[,1],
    pca$x[,2],
    col = as.numeric(data$Species),
    pch = 19,
    xlab = "Principal Component 1",
    ylab = "Principal Component 2",
    main = "PCA of Iris Dataset")
```

```
legend("topright",
       legend = levels(data$Species),
       col = 1:3,
       pch = 19)
```



t-SNE

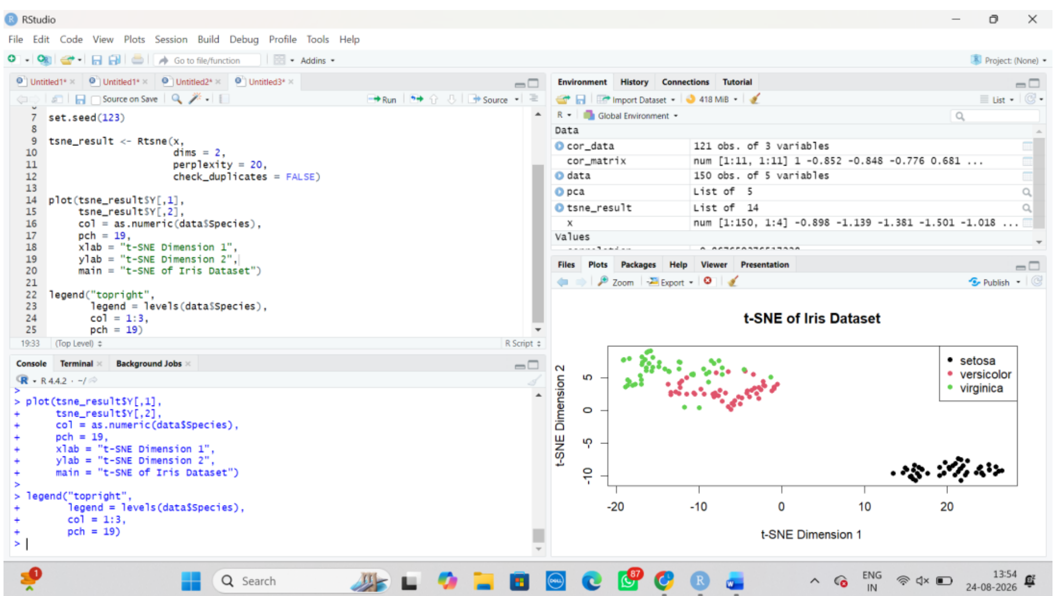
First install the package once:

```
install.packages("Rtsne")
```

Then:

```
library(Rtsne)

data <- iris

x <- scale(data[, 1:4])

set.seed(123)
```

```
tsne_result <- Rtsne(x,
                dims = 2,
                perplexity = 20,
                check_duplicates = FALSE)

plot(tsne_result$Y[,1],
    tsne_result$Y[,2],
    col = as.numeric(data$Species),
    pch = 19,
    xlab = "t-SNE Dimension 1",
    ylab = "t-SNE Dimension 2",
    main = "t-SNE of Iris Dataset")

legend("topright",
      legend = levels(data$Species),
      col = 1:3,
      pch = 19)
```



Simple conclusion

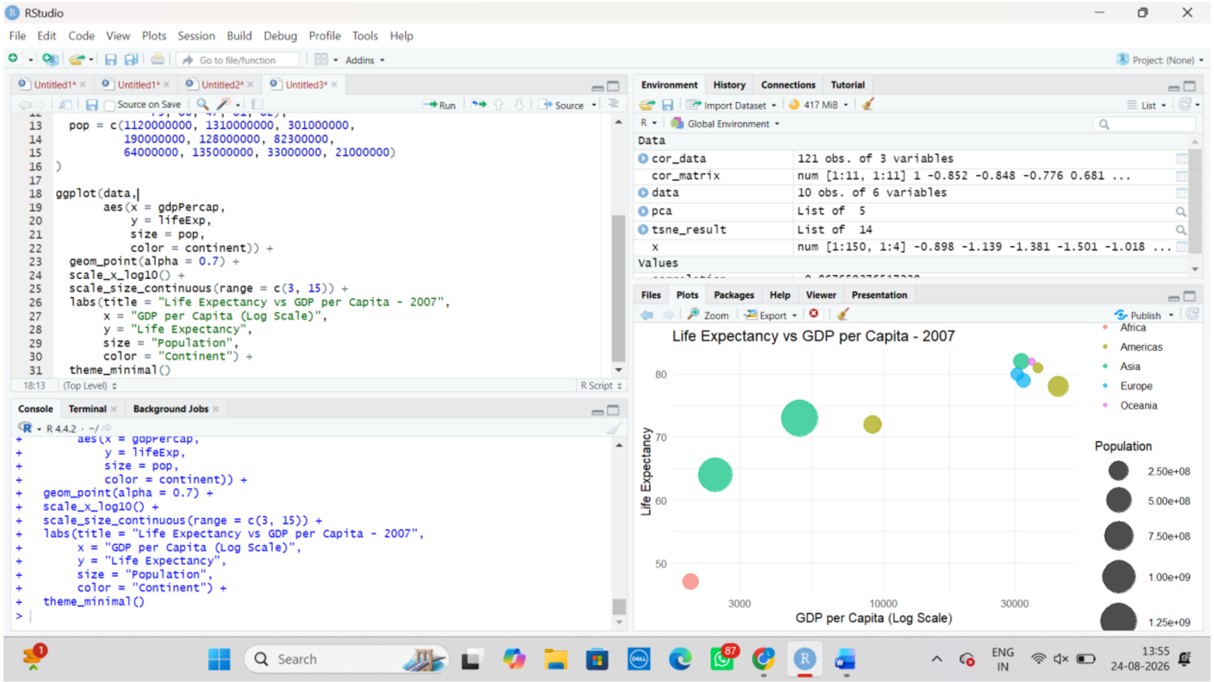
PCA is suitable when we want to reduce dimensions while preserving maximum variance and retaining interpretable components. **t-SNE** is more suitable for visualizing complex, non-linear structures and identifying clusters in high-dimensional data.

Q4. Bubble Chart for Multivariate Paired Data

The question asks for the 2007 data with GDP per capita on the x-axis, life expectancy on the y-axis, population as bubble size, and continent as colour.

R Code

```
library(ggplot2)


data <- read.csv("gapminder2007_practice.csv")


data2007 <- subset(data, year == 2007)


ggplot(data2007,
       aes(x = gdpPercap,
           y = lifeExp,
           size = pop,
           color = continent)) +
  geom_point(alpha = 0.7) +
  scale_x_log10() +
  scale_size_continuous(range = c(2, 15)) +
  labs(title = "Life Expectancy vs GDP per Capita - 2007",
       x = "GDP per Capita (log scale)",
       y = "Life Expectancy",
       size = "Population",
       color = "Continent") +
  theme_minimal()
```



Interpretation

A visible pattern is that countries with **higher GDP per capita generally have higher life expectancy**. Large bubbles represent countries with larger populations, while countries with relatively low GDP and life expectancy form a lower-left cluster.

Conclusion

The bubble chart effectively represents **four variables simultaneously**:

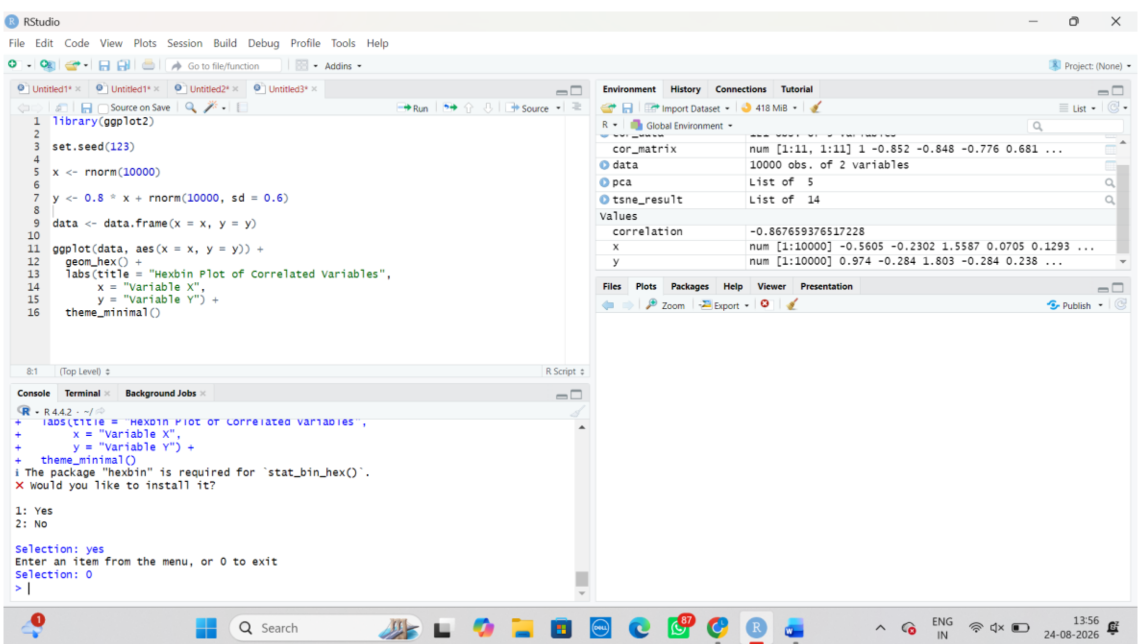
- X-axis → GDP per capita
- Y-axis → Life expectancy
- Bubble size → Population
- Colour → Continent

Q5(a). Hexbin Plot

R Code

```
library(ggplot2)

library(MASS)


set.seed(123)
```

```
data <- mvrnorm(
  n = 10000,
  mu = c(0, 0),
  Sigma = matrix(c(1, 0.8,
                   0.8, 1), nrow = 2)
)

df <- data.frame(x = data[,1], y = data[,2])

ggplot(df, aes(x = x, y = y)) +
  geom_hex() +
  labs(title = "Hexbin Plot of Correlated Variables",
       x = "Variable X",
       y = "Variable Y") +
  theme_minimal()
```



Why Hexbin instead of a normal scatterplot?

A standard scatterplot becomes difficult to interpret when **10,000 observations** are plotted because many points overlap. A hexbin plot groups observations into hexagonal bins and uses the density of each bin to show where observations are concentrated.

Conclusion

Hexbin plots are preferable for large datasets because they reduce overplotting and clearly show the density and distribution of observations.

Q5(b). Network Graph

The PDF specifies that network_edges.csv contains **15 weighted edges and 8 nodes**: Ava, Ben, Cleo, Dax, Elin, Fenn, Grey and Hana.

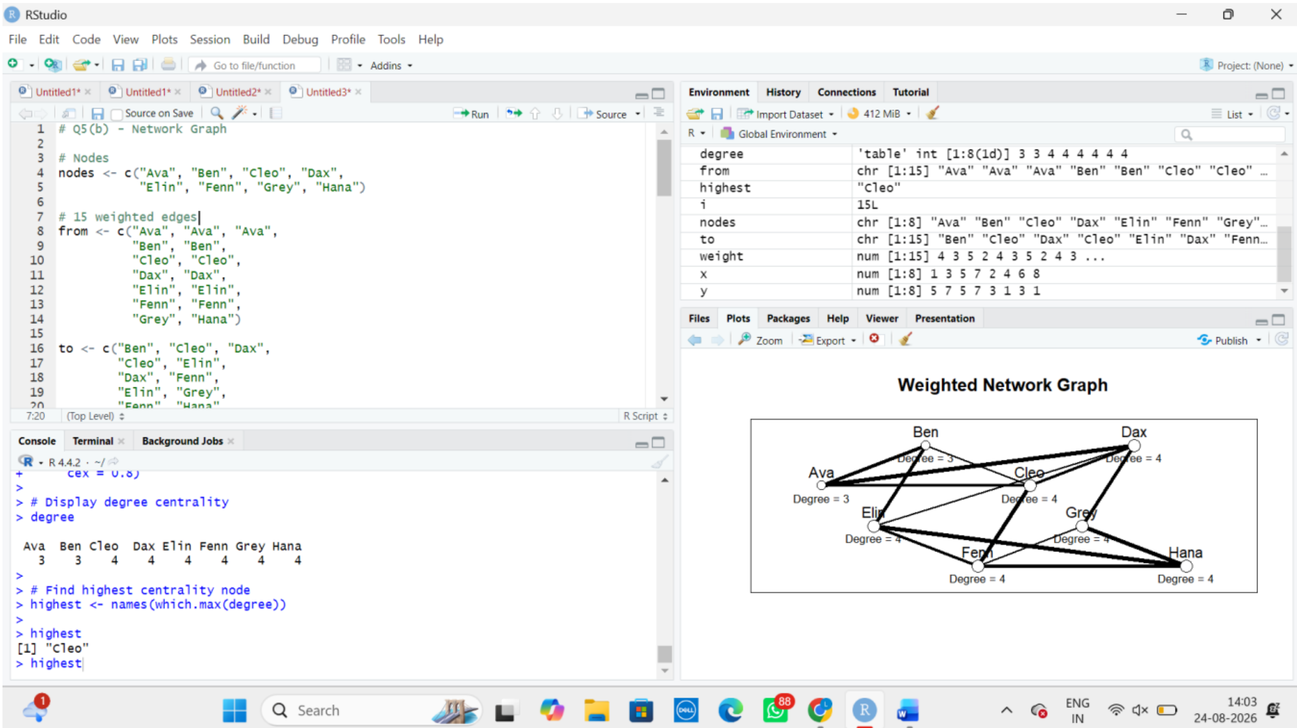
R Code

```
library(igraph)


edges <- read.csv("network_edges.csv")


g <- graph_from_data_frame(
  edges,
  directed = FALSE
)


degree_values <- degree(g)


plot(g,
     vertex.size = degree_values * 8,
     vertex.label.color = "black",
     edge.width = edges$weight)


degree_values
```

To find the highest-centrality node

```
highest <- names(which.max(degree_values))
```

highest



Interpretation

The node with the **highest degree centrality** is the most strongly connected node in the network. It can be considered an important hub because it has direct connections with more nodes than the other nodes.